

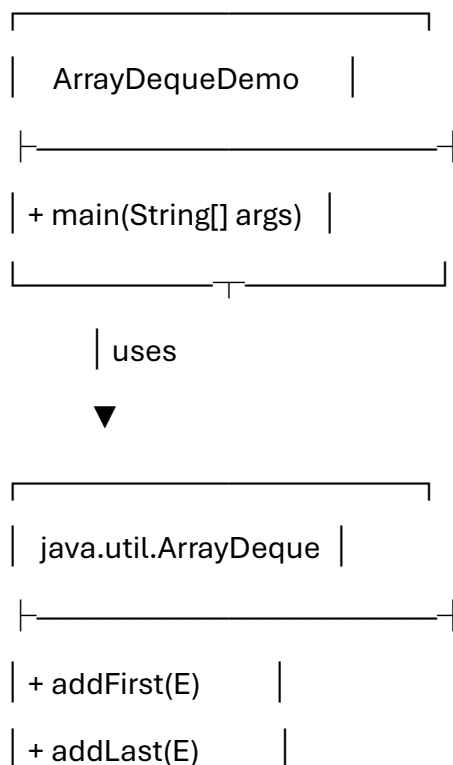
Aim

To write a Java program to demonstrate the important methods of the **ArrayDeque** class, such as `addFirst()`, `addLast()`, `offerFirst()`, `offerLast()`, `pollFirst()`, `pollLast()`, `peekFirst()`, and `peekLast()`.

Algorithm

1. Start the program.
2. Import the `java.util.ArrayDeque` class.
3. Create an `ArrayDeque` object.
4. Insert elements at the front and end using `addFirst()` and `addLast()`.
5. Insert elements using `offerFirst()` and `offerLast()`.
6. Display the deque.
7. Use `peekFirst()` and `peekLast()` to view the first and last elements.
8. Use `pollFirst()` and `pollLast()` to retrieve and remove the first and last elements.
9. Display all results.
10. Stop the program.

UML Diagram



+ offerFirst(E)	
+ offerLast(E)	
+ pollFirst()	
+ pollLast()	
+ peekFirst()	
+ peekLast()	

Code:

```
package Observations;
```

```
import java.util.ArrayDeque;
```

```
public class ArrayDequeClass {
```

```
    public static void main(String[] args) {
```

```
        // Creating ArrayDeque
```

```
        ArrayDeque<Integer> deque = new ArrayDeque<>();
```

```
        // addFirst() and addLast()
```

```
        deque.addFirst(20);
```

```
        deque.addLast(30);
```

```
        deque.addFirst(10);
```

```
        System.out.println("After addFirst() and addLast(): " + deque);
```

```
        // offerFirst() and offerLast()
```

```
        deque.offerFirst(5);
```

```
        deque.offerLast(40);
```

```
System.out.println("After offerFirst() and offerLast(): " + deque);

// peekFirst() and peekLast()
System.out.println("First element using peekFirst(): "
    + deque.peekFirst());
System.out.println("Last element using peekLast(): "
    + deque.peekLast());

// pollFirst()
System.out.println("Removed first element using pollFirst(): "
    + deque.pollFirst());

System.out.println("Deque after pollFirst(): " + deque);

// pollLast()
System.out.println("Removed last element using pollLast(): "
    + deque.pollLast());

System.out.println("Deque after pollLast(): " + deque);
}
}
```

Output:

```
es' -cp' 'C:\Users\Sahithi_pyla\AppData\Roaming\Code\User\work
bservations.ArrayDequeClass'
After addFirst() and addLast(): [10, 20, 30]
After offerFirst() and offerLast(): [5, 10, 20, 30, 40]
First element using peekFirst(): 5
Last element using peekLast(): 40
Removed first element using pollFirst(): 5
Deque after pollFirst(): [10, 20, 30, 40]
Removed last element using pollLast(): 40
Deque after pollLast(): [10, 20, 30]
PS C:\Users\Sahithi_pyla\Desktop\Sahithi_java>
```

Result:

Thus, the Java program successfully demonstrates the ArrayDeque methods addFirst(), addLast(), offerFirst(), offerLast(), pollFirst(), pollLast(), peekFirst(), and peekLast().